$\Lambda \to p\pi^-$ in the ePIC B0 tracker

Acceptance, tracking efficiency and the reconstructed Λ mass
old B0 geometry vs eic/epic#1194

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Code, results, report: `https://github.com/gadininio/eic-B0-detector`

Introduction and motivation

- Far-forward Λ: production and polarization in the target fragmentation region, Λ tagging in exclusive/diffractive reactions
- At high energy the Λ flies at a few to $\sim$20 mrad from the proton beam and decays **metres downstream**: $c\tau = 7.89\,\text{cm} \Rightarrow \langle L \rangle \approx 5.3\,\text{m}$ at 75 GeV
- The B0 tracker sits at $z \approx 5.8$–$6.9\,\text{m}$, inside the B0 dipole
- $\Lambda \to p\pi^-$ (64 %) must be reconstructed in B0 from tracks that **do not come from the IP**

Goal: reconstruct the Λ mass from B0 tracks, step by step, and understand each loss:

- acceptance (what the detector covers) $\to$ tracking efficiency $\to$ acceptance $\times$ efficiency $\to$ mass
- compare the B0 geometry of the nightly with the realistic B0 of **eic/epic#1194**

Setup

Sample (single-Λ gun)

- $p = 50$–$100\,\text{GeV}$, $\theta = 8$–$18\,\text{mrad}$, ϕ uniform
- 25 mrad crossing angle, 18×275 craterlake
- 100 ev. (seed 42), **1000 ev.** (2×500) per geometry
- 5000 geantinos per geometry

Composition (1000 ev.)

- $\Lambda \to p\pi^-$ 48–50 %, $n\pi^0$ 25–28 %, other 24–25 % (interactions)
- 88–90 % of $p\pi^-$ decays before B0
- $M(p\pi^-)$ from true momenta $= 1.115683\,\text{GeV}$

Software

- ddsim in eic-shell (nightly)
- **old** geometry: epic `43f427e`, sensors 300 μm
- **new** geometry: epic main `ec774d8` (#1194), sensors 50 μm
- EICrecon main `8b44cae` (with #2901)
- ACTS CKF with **truth seeding**

Definitions

$\Lambda \to p\pi^-$ decay	generated Λ with exactly two daughters, p and π^-
Accepted daughter	its own *simulated* B0 hits cover ≥ 3 layers (enough for a track)
Reconstructed daughter	a reconstructed particle whose best truth match is the daughter
Geometric acceptance	straight lines from the IP at the Λ angles crossing ≥ 3 layers (geantinos). *Not* a Λ acceptance: the Λ is neutral
Daughter acceptance A	accepted daughters / $p\pi^-$ decays (decay point, field, material; no detector response)
Tracking efficiency ε	reconstructed accepted daughters / accepted daughters (geometry cancels)
Acceptance $\times$ **efficiency**	reconstructed daughters / $p\pi^-$ decays $= A \cdot \varepsilon$
Purity	signal / selected candidates (truth only labels)
B0 threshold	energy per pixel needed for a hit (EICrecon default 10 keV)

Angles w.r.t. the proton beam (crossing-angle rotation undone). Layers: hits more than 50 mm apart in z.

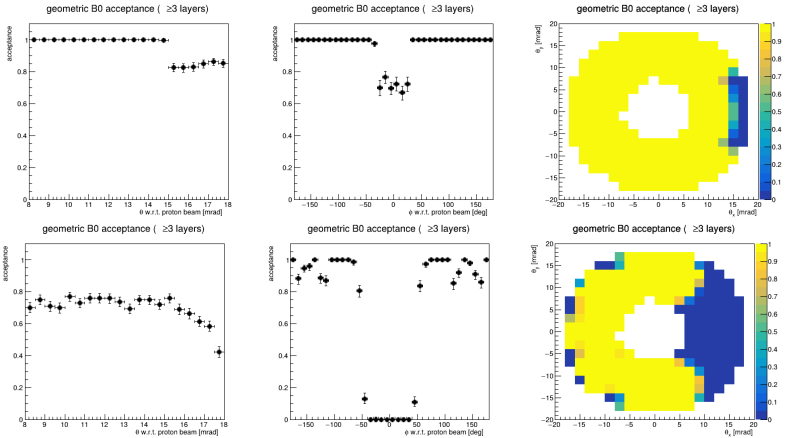
First look: only 4 B0 tracks per 100 Λ

- Nightly EICrecon (`a7bab3f`): the truth seeder only seeded particles from $|z| < 200\,\text{mm} \Rightarrow$ Λ daughters from decays metres downstream had no seed
- **EICrecon#2901** (in main): dedicated `B0TrackerTruthSeeds`, vertex $z < 6\,\text{m}$, perigee at the true vertex

100 events: nightly → main	old geometry	new geometry
B0 reconstructed hits	1416	593
truth seeds	8 → 650	4 → 504
B0 tracks	4 → **77**	1 → **44**
events with a + and a − track	1 → 18	0 → 14

- Same hits: the whole gain is seeding. Seeds include shower particles ⇒ truth matching needed

Geometric acceptance along the Λ direction (geantinos)



- ≥ 3 layers: **95.3 %** (old, top) vs **70.1 %** (new, bottom)
- New geometry: **gap of** $\pm 45^\circ$ **in** ϕ on the electron-beam side; losses above 16 mrad

Decay kinematics: a hard proton and a soft π^-

- $Q = 38\,\text{MeV}$: in the Λ frame $p^* = 0.101\,\text{GeV}$, $E_p^* = 0.943\,\text{GeV}$, $E_\pi^* = 0.172\,\text{GeV}$
- Both daughters move with $\approx$ the Λ velocity and share its momentum as their rest-frame energies:

$$\frac{p_p}{p_\Lambda} \in [0.76, 0.94] \;\Rightarrow\; 38\text{–}94\,\text{GeV}, \qquad \frac{p_\pi}{p_\Lambda} \in [0.06, 0.25] \;\Rightarrow\; 3\text{–}25\,\text{GeV}$$

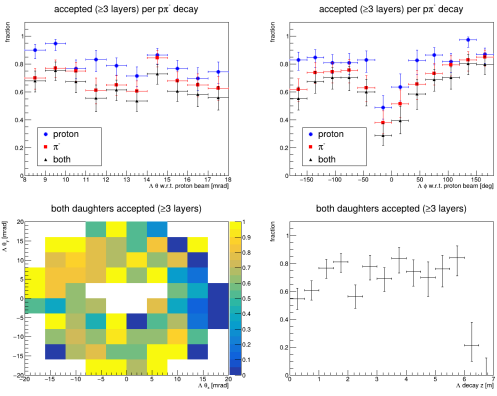
- The soft π^- is bent much more by the B0 dipole
- Momentum ratio, independent of p_Λ:

$$r = \frac{p_p}{p_\pi} = \frac{E_p^* + p^* \cos\theta^*}{E_\pi^* - p^* \cos\theta^*} \in [3.09,\ 14.6]$$

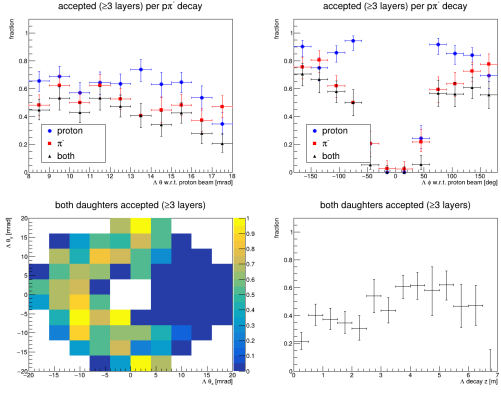
(used later for a selection without truth)

Acceptance for the Λ daughters (full simulation, simulated hits)

old geometry



new geometry

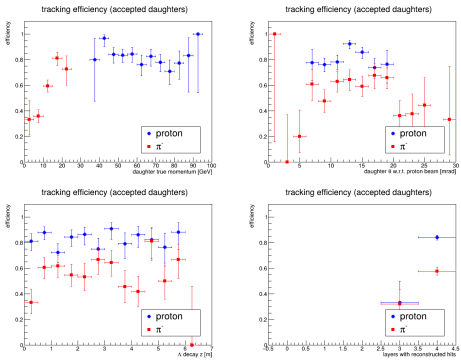


per $p\pi^-$ decay, ≥ 3 layers	proton	π^-	**both**
old	80 %	69 %	**63 %**
new	61 %	49 %	**41 %**

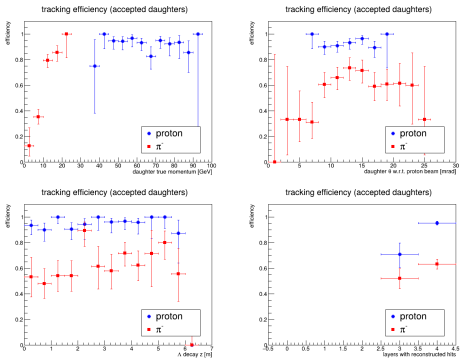
$\approx$ 30 points below the geometric number: decay point, opening angle, π^- bending. Rises with the decay distance.

Tracking efficiency for accepted daughters

old geometry



new geometry, 5 keV



- proton 82–93 %, π^- 56–61 %. **The π^- loss is a momentum effect:** 12–36 % below 10 GeV, 75–90 % at 10–25 GeV (as good as the proton)

Where are the daughters lost?

accepted daughters, proton / π^-	old	new, 10 keV	new, 5 keV
accepted	404 / 349	295 / 240	295 / 240
rec. hits in ≥ 3 layers	100 / 100 %	93 / 93 %	100 / 100 %
seed	99 / 99 %	97 / 96 %	97 / 96 %
CKF track	82 / 56 %	86 / 58 %	93 / 61 %
filtered track	82 / 56 %	86 / 58 %	93 / 61 %
reconstructed particle	82 / 56 %	86 / 58 %	93 / 61 %

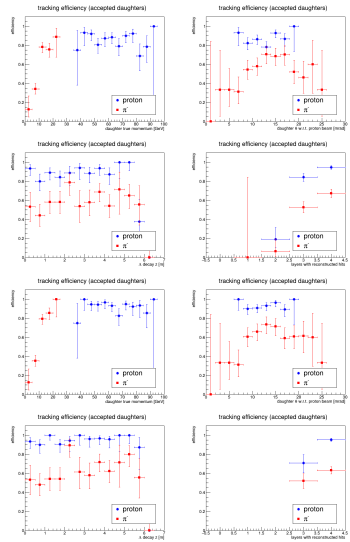
- Seeds exist; the loss is in the **CKF** (seed $\rightarrow$ track); filtering and particle building lose nothing
- **Seed momentum smearing (10 %) ruled out:** rerun with `MomentumSmear=0` $\Rightarrow$ 71 vs 70 of 120 π^-
- Open: CKF χ^2/measurement cuts? multiple scattering? (0,0,0) perigee extrapolation of #2901?

The B0 threshold: 50 μm sensors need 5 keV

per hit (hadrons)	old	new
sensor thickness	300 μm	50 μm
most probable deposit	95 keV	**13.8 keV**
below 10 keV	2.6 %	9.3 %
below 5 keV	1.6 %	0.3 %

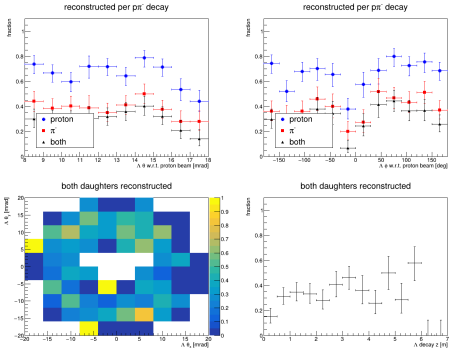
- MIP: $\approx 0.28\,\text{keV}/\mu\text{m}$ in Si
- 10 keV cuts the Landau tail: 7 % of accepted daughters lose layers
- 5 keV recovers them; proton eff. 86 $\rightarrow$ 93 %
- `-PB0TRK:B0TrackerRawHits:threshold=5e-6`
- Not the EICrecon default yet

new geometry: 10 keV (top) / 5 keV (bottom)

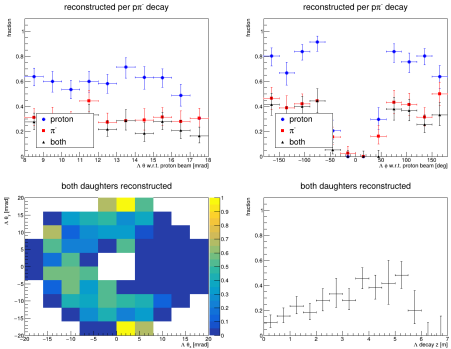


Acceptance × efficiency

old geometry

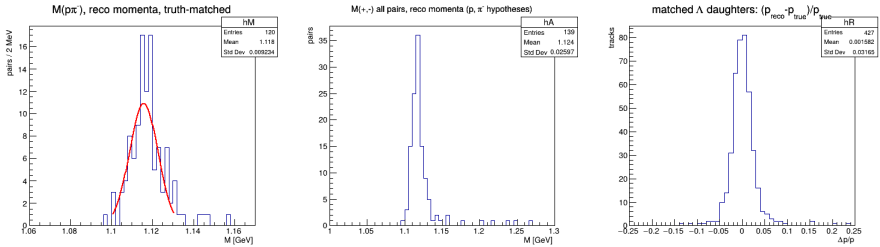


new geometry, 5 keV



proton / π^- / both	old	new, 10 keV	new, 5 keV
A	80 / 69 / 63 %	61 / 49 / 41 %	61 / 49 / 41 %
ε	82 / 56 / 47 %	86 / 58 / 50 %	93 / 61 / 59 %
$A \times \varepsilon$	66 / 39 / **30 %**	53 / 29 / **21 %**	57 / 31 / **25 %**
per generated Λ (both)	**15.0 %**	**10.1 %**	**12.0 %**

Λ mass: truth-matched pairs

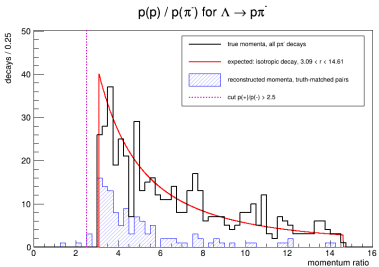


new geometry, 5 keV: truth-matched pairs (fit), all $(+,-)$ pairs, $\Delta p/p$ of the daughters

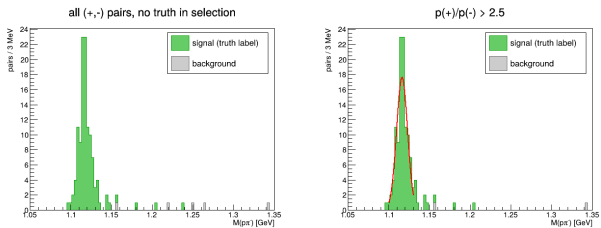
	old	new, 10 keV	new, 5 keV
pairs (of 1000 Λ)	150	101	120
mean [GeV]	1.1177 ± 0.0007	1.1160 ± 0.0010	1.1159 ± 0.0008
σ [MeV]	6.3 ± 0.6	7.4 ± 1.1	6.9 ± 1.1

PDG 1.1157 GeV. Old geometry +2 MeV ($\approx 3\sigma$), not understood.

Λ mass without truth in the selection



true ratio 3.09–14.58, as predicted; cut at 2.5 loses 2 % of the signal



all $(+,-)$ pairs (left), after $p(+)/p(-) > 2.5$ (right); green = signal (truth label)

	old	new, 10 keV	new, 5 keV
after the cut: candidates / **purity**	155 / **95 %**	103 / **96 %**	122 / **97 %**
±20 MeV window: candidates / signal	144 / 142	92 / 92	110 / 110

Tracks still truth-seeded; single-Λ events.

Summary

- B0 tracks for displaced Λ daughters need the B0 truth seeding of **EICrecon#2901**: 4 $\rightarrow$ 77 tracks / 100 Λ
- Realistic B0 (**eic/epic#1194**): $\pm 45^\circ$ ϕ gap on the electron side
  - geometric acceptance 95 $\rightarrow$ 70 %; both daughters accepted 63 $\rightarrow$ 41 %
- 50 μm sensors need a **5 keV threshold** (10 keV loses 7 % of accepted daughters)
- Tracking efficiency: proton 82–93 %, π^- 56–61 %; **soft tracks ($<$ 10 GeV) lost in the CKF**, not due to seed smearing
- Both daughters reconstructed: **15 %** (old), **10 %** (new, 10 keV), **12 %** (new, 5 keV) of all Λ
- $M(p\pi^-)$ at the PDG value, $\sigma \approx$ 6–7 MeV; one kinematic cut gives $\approx$ 97 % purity without truth

Next: CKF debugging for soft tracks; realistic seeding (EICrecon#3028); cross-check with EICrecon#2926 (ChargedLambdaReconstruction); full events with backgrounds and vertexing

Backup: reproducing the results

Repository `https://github.com/gadininio/eic-B0-detector`

- `macros/acceptance/`: `b0_geometric_acceptance.C`, `lambda_daughter_acceptance.C`
- `macros/efficiency/`: `lambda_tracking_efficiency.C`, `lambda_acceptance_x_efficiency.C`
- `macros/mass/`: truth, truth-matched, no-truth mass, momentum-ratio check
- `scripts/production/lambda_sample.sh`: simulation + reconstruction (10 and 5 keV) + copy to EOS
- `scripts/make_results.sh`: every plot and number in `results/`

Pitfalls

- EOS not visible in eic-shell; parallel jobs in one folder race on `calibrations/`
- ddsim drops hits below 1 keV (geantinos!): `--filter.tracker ""`
- EICrecon parameters need the plugin prefix (`tracking:`, `B0TRK:`)